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Effective Diagnosis of Alzheimer's Disease using Modified Decision Tree Classifier

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Abstract

Alzheimer's disease (AD) is described as a severe form of the neural disorder that collectively degenerate the essential cognitive activities of a human being (thinking, memory retention, etc.) in particular among the elderly individuals and eventually results in death. In addition to the adverse ill-health effects on the patients, AD imposes paramount responsibility and burden on the caretakers too. Several genetic and pathological traits and non-invasive diagnostic strategies are being vigorously investigated and explored to discover the early onset of this debilitating disease. The prognosis of AD assumes importance, as the deterioration of health due to its progression may be either contained or controlled. Moreover, early and accurate detection of AD helps medical practitioners to prescribe case-specific medical treatment procedure. Among the popular machine learning algorithms, decision tree technique is widely used for classification/prediction, due to its accuracy and speed. This research article presents a novel decision tree-based classification technique, with optimum hyper parameter tuning, that is ideally suitable for AD diagnosis, even at the early stages of development. The performance of this newly proposed Decision Tree Classifier with Hyper Parameters Tuning (DTC-HPT) is validated on the Open Access Imaging Studies Series (OASIS) dataset that contains patients' data on the different stages of AD. The DTC-HPT is designed with the primary objective to classify the nature of brain abnormality using the most relevant and potentially significant data attributes/parameters. The efficiency of DTC-HPT on AD classification is measured as Accuracy, Precision, Recall, and F1-Score. The correctness of AD classification by DTC-HPT with an average accuracy of 99.10% endorses that this classification technique can be used for AD detection on the AD clinical datasets.

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Keywords: Alzheimer's Disease, Feature Selection, Classification, Decision Tree, Early Detection, Hyper Parameter Tuning
1. INTRODUCTION

Machine learning (ML) - a branch of artificial intelligence is widely explored in health care practices as a technological tool for the validation and verification of manual diagnosis. The availability of voluminous clinical data, emergence of new computational models and high performance computers have motivated the researchers to continuously experiment on the machine learning algorithms to solve the problems in the domain of medicine.

For instance, the diagnosis of post-concussion syndrome (PCS) that has many similarities to AD is done with Computer Assisted Diagnosis (CAD) systems. Such diagnostic systems are primarily anchored on the principles of statistical, probabilistic and optimization techniques, which assure greater accuracy than that of traditional computational models. Researchers continue to experiment with machine learning techniques to broader boundaries into medicine for accurate diagnosis, classification, analysis and prediction. It is also evident that the accuracy of /classification of clinical data is duly affected by various elements of processing such as data imputation and selection of features/attributes. Intending to provide feasible solutions for these issues, this research article proposes a new mechanism termed as Decision Tree Classifier with Hyper Parameters Tuning (DTC-HPT) for the precise classification of AD from clinical Alzheimer's datasets. Further, this model-based method gives proper inference for early diagnosis of AD and distinguishes between AD and a healthy brain. The vital attributes of the datasets for AD diagnosis are: Gender, Age, Hand (right/left brain hemisphere), Educ (education), SES (socio-economic status), Mini-Mental State Examination (MMSE), CDR (Clinical Dementia Rating), eTIV (estimate total intracranial volume), nWBV (normalized whole brain volume), Group (stage of AD), ASF (Atlas Scale Factor), Visit (number of visit) and MR Delay. A decision tree is constructed by the selection of root node, non-leaf nodes and leaf nodes from among the attributes based on their significance computed as Entropy, Information gain and Gini Index. The DTC-HPT approach is observed to produce accuracy of 92% and above.

Early diagnosis of AD enables us to propose suitable therapeutic solutions for the subjects by the practitioners.

Magnetic Resonance Imaging (MRI) is an appropriate diagnostic technique that provides high-resolution images and 1877-0509 © 2019 The Authors. Published by Elsevier B.V.

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contrasts to the brain's grey and white matter. MRI acts as a reliable source for medical image analysis for the prognosis of AD and Dementia cases [1]. This article presents the procedure of DTC-HPT for AD diagnosis and its merits.

The remaining sections of this paper are organized as follows: the related literature is discussed in Section II. Section III describes the methodology of the DTC-HPT. The dataset description, and the experimental outcomes are given in section IV. Section V describes the merits of the experimental outcomes and the conclusion on this research work is given in Section VI.

II. Related Work

Javier Escudero *et al.* reported on the diagnosis of Alzheimer's Disease, using machine learning techniques [2]. Roman Filipovych *et al.* used a supervised approach of Support Vector Machine (SVM) model to classify images [3]. Ali Hamou *et al.* [4] proposed a cluster analysis technique for AD diagnosis from Magnetic Resonance Brain Images in the light of the selection tree. Xia-an Bi *et al.* [5] proposed the random neural system group to enhance the order of execution. The dataset for their experimentation was selected from the Alzheimer's Disease Neuroimaging Initiative (ADNI). The Elman Neural Network was confirmed to be an ideal base classifier that utilizes the arbitrary neural system cluster that is dependent on the outcome of highlighted selections, with 92.31% of accuracy

Bi *et al.* [6] used on the random support vector machine clustering approach to diagnose AD and to group the disease regions into inferior frontal gyrus, superior frontal gyrus, precentral gyrus and cingulate cortex. Sandhya Joshi *et al.* [7] developed multiple models for the classification of AD using various machine learning techniques, namely multilayer perception, bagging, decision tree, CANFIS (Co-Active Neuro-Fuzzy Inference System) and genetic algorithm. The CANFIS method's classification precision was 99.55%. Claudia Plant *et al.*, [8] developed a hybrid model using SVM and Bayesian Classifier to detect the brain atrophy patterns as well as to predict AD. A pattern-matching index of 92% was obtained for their method. Luis *et al.* [9] developed a new methodology for the classification of MR brain images into normal and AD affected images. Their method is underlined by MRI feature extraction in the wavelet domain followed by dimensionality reduction and SVM classification.

Li *et al.* [10] proposed a current strategy based on the Principle Component Analysis (PCA). It utilizes models of continuous selection and dropout as well as the model of Restricted Boltzmann Machine (RBM), which is a profound teaching method. Ye *et al.* [11] created a technique to analyze AD from medical images using ML-based multimodal data fusion.

Rama *et al.* [12] had designed an Integrated Visualization Management (IVM)-based multiclass categorization. In this method, features from structural MRI, is utilized along with a logistic regression method. This model is used to reduce the computational cost. Based on the features extracted from the Region of Interest (RoI) for training and testing, a maximum accuracy of 70% was obtained on Healthy Control (HC) / AD images.

III. Decision Tree Classifier Hyper Parameters Tuning (DTC-HPT)

Decision tree classification is a traditional data classification technique which has several applications across the disciplines. Decision trees are constructed in numerous ways as branch-like fragments. The decision tree starts with a root node; leaf nodes denote the class labels, whilst the intermediate nodes are referred to as non-leaf nodes. The data attribute with the highest priority in decision making is chosen as the root node. The splitting process of a decision tree is decided upon the data values of the respective node [13]. In this method, four vital attributes contribute to a higher probability of detecting AD. They are Gender (G), Age (A), Hand (H), Mini-Mental State Examination (MMSE), and Clinical Dementia Ratio (CDR). These attributes represent the critical factors that have a bearing on the presence and development of AD. In machine learning, the decision tree classifier learns during the training phase and the effectiveness of the classifier is assessed during the testing phase. The decision tree depth and the distribution of training and test information dynamically influence the performance and effectiveness of the classifier. The hyperparameters for the Decision tree classifier are listed herein under [14].

A. max: int or None (default = None), optional

The tree node's maximum depth is extended to cover all sheets pure or to allow all sheets to contain less than min samples split samples.

B. Criterion: (default="gini") "string option.

The estimation function of a divided quality. The criteria supported are "gini" for the Gini impurity and "entropy."

C. splitter :string, optional (default="best")

The strategy used for dividing every node. Strategies supported are "best" to choose the best division.

D. min samples split: int, float. (default=2) optional.

The minimum amount for dividing an inner node: It works in two modes namely int and float

E. None of int, optional (default = None) max Leaf nodes

Develop a tree in the best fashion with max leaf nodes. For the best nodes, the comparative impurity decreases. If None, the number of nodes will be unlimited.

F. max features: int, float, string, none (default = None) optional

- The number of functions to consider when searching for the best division.
- If int, consider the max features at each division.
- If float is a fraction of max features and int(max features*n features) is taken into account in every division.
If "auto", then max_features=sqrt(n_features).
- MAX features = sqrt(n features) if "sqrt"
- If "log2," max features = log₂(n features) will be used.
- If none, then max function = n functionality.

Methodology of DTC-HPT

The block diagram of DTC-HPT Classifier Model is shown in Fig.1.

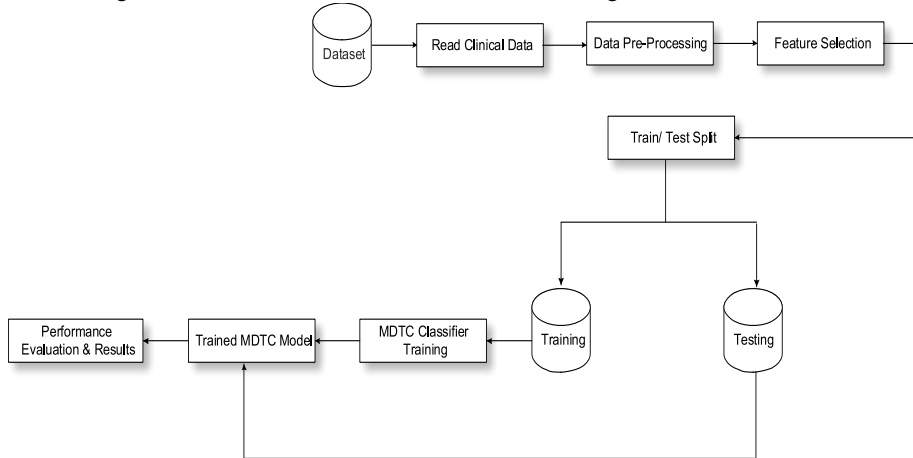


Fig. 1: Block Diagram of DTC-HPT Classifier Model

Algorithmic Description of DTC-HPT (Proposed)

Input: Clinical Data from MRI Brain images

Output: Classification of Alzheimer's disease Stages

Phase I: Pre-Processing

- Step 1: Read the clinical data
- Step 2: Pre-Processing: Data Imputation for missing values
- Step 3: Perform One-hot Encoding

Phase II: Classification

- Step 4: Split data for Training and Testing
- Step 5: Apply DTC-HPT classifier on training dataset
- Step 6: Classifier algorithm computes the information of D as

$$\text{info}(D) = -\sum_{i=1}^m p_i \log_2(p_i) \quad (1)$$

where p_i is the probability of prediction groups, i .

- Step 7: The information gain of imputed dataset, DI is computed as

$$\text{info}_x(DI) = \sum_{j=1}^r \frac{|D_j|}{|D|} \times I(DI_j) \quad (2)$$

where information gain I is an attribute of DI

$$\text{Gain}(X) = \text{Info}(D) - \text{Info}_x(DI) \quad (3)$$

(The attribute that gives us the highest gain is used as the splitting attribute.)

- Step 8: Evaluate the Performance DTC-HPT on test dataset.

- Step 9: Stop.

IV. Results and Discussion

Dataset Description

The OASIS dataset is a compilation of MRIs from 416 topics aged 18 to 96 years. The MRI collections are the cross-sectional pictures of older adults, young and middle-aged with non-demented and demented. The topics concerned come from both sexes and are right-handed. Of the 416 topics, 100 subjects over 60 years of age were clinically diagnosed with Alzheimer's disease (AD) between very mild to moderate. The dataset also comprises of 150 non-demented and demented elderly adults aged 60 to 96 with longitudinal MRI information. Each of these patients was scanned for 373 image sessions on two or more visits with a gap of at least one year. Of these, 72 participants were classified as non-demented throughout the research; at the first visit, 64 were classified as demented, out of which 51 were classified from mild to moderate AD. Finally, 14 subjects from this collection were considered as non-demented at their first visit and were characterized as demented at a later visit. The attributes description of the dataset is given in Table 1[15].

Performance Evaluation Metrics

The evaluation metrics bring out the merits of a classifier model based on the performance assessment and analysis [16]. The confusion matrix comprises of the following components:

- **True Positives (TP):** Classifier predicted the test samples as diseased when they have the disease.
- **True Negatives (TN):** Classifier predicted no disease when they don't have.
- **False Positives (FP):** Classifier predicted the test samples as diseased; they don't have.
- **False Negatives (FN):** Classifier predicted no disease, but they have.

The final table of the confusion matrix would contain the average values for all classes combined.

The F1-Score

This score is a metric that evaluates the performance of a classifier, also known as the F-Score or F-Measure. The F1-Score is computed using both the precision of the classifier P and the recall R, where P is the amount of true positive results between all the positive results determined by the classifier and R is the amount of true positive results among all appropriate samples [17]. The F1-score is the most accurate and harmonic mean to remember. Greater the F1-Score, the greater the P and R values

Table 1: OASIS Dataset Attributes (Demographic & Clinical Information)

Demographic Information		Clinical Information	
Descriptor	Description	Descriptor	Description
M/F	Gender	MMSE	Mini-Mental State Examination score [Range: 0 (Worst) to 30 (Best)]
Hand	Handedness (Right-Handed / Left -Handed)		
Age	Age in years		
EDUC	Years of education	CDR	Clinical Dementia [AD Rating: 0 (No dementia); 0.5 (Very Mild); 1 (Mild); 2 (Moderate)]
SES	Socioeconomic status Categories: 1 (Highest status) to 5 (Lowest status)		

The classifiers were trained using the clinical dataset, and the performance metrics including precision, accuracy, recall, F1-Score were determined for each classifier. The results presented here will provide a comparison of the performance of the Decision Tree classifiers concerning each metric.

Table 2: Performance of DTC without Data Imputation

Group	Precision	Recall	F1-Score	Support
Mild AD	0.13	0.17	0.15	12
Moderate AD	0.00	0.00	0.00	2
Non-demented	0.54	0.63	0.58	59
Very Mild AD	0.29	0.21	0.24	39
Classification Report				
Macro Avg	0.24	0.25	0.24	112
Weighted Avg	0.40	0.42	0.40	112
Accuracy: 0.41964285714285715 (41.96%)				

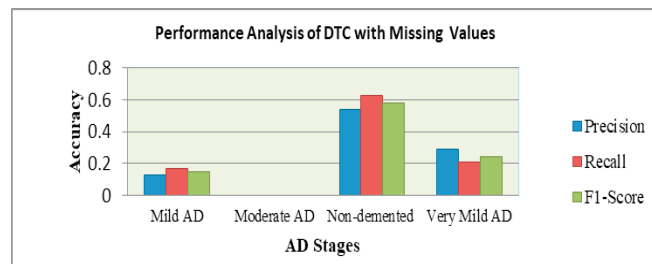


Fig.2: Performance Analysis of DTC with Missing Values

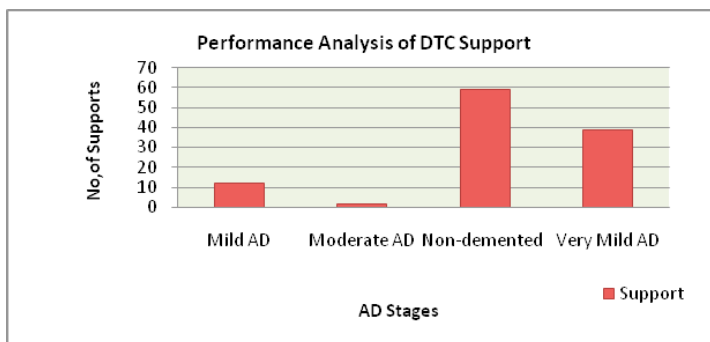


Fig. 3: Performance Analysis of DTC Support

Table 2 and Fig.2, Fig. 3 depict the performance of Decision Tree Classifier, without data imputation for the missing data in terms of the chosen metrics. The accuracy of the model is 41.96%. Thus, table shows that the performance of DTC is very low, prior to imputation.

Table 3: Performance of DTC-HPT (with Data Imputation)

Group	Precision	Recall	F1-Score	Support
Mild AD	0.89	1.00	0.94	8
Moderate AD	0.00	0.00	0.00	1
Non-demented	1.00	1.00	1.00	63
Very Mild AD	1.00	1.00	1.00	40
Classification Report				
Macro Avg	0.72	0.75	0.74	112
Weighted Avg	0.98	0.99	0.99	112
Accuracy: 0.9910714285714286 (99.10%)				

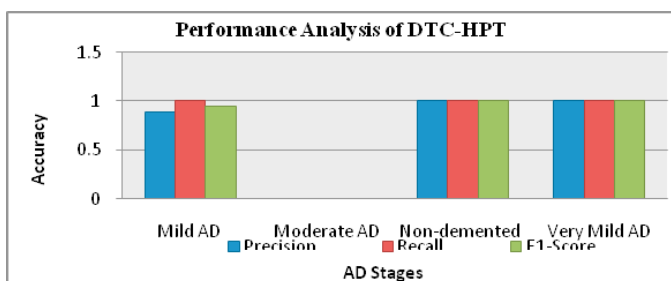


Fig. 4: Performance Analysis of DTC-HPT

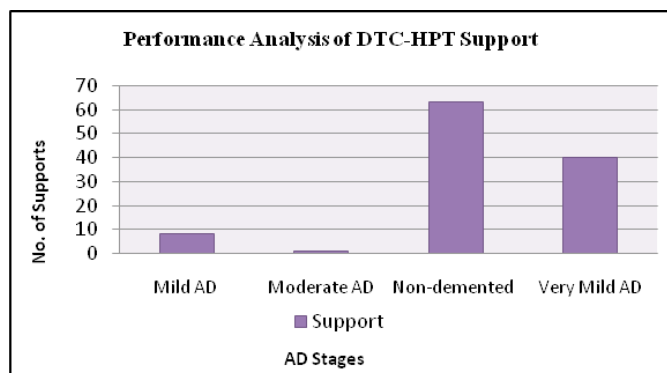


Fig. 5: Performance Analysis of DTC-HPT Support

Fig. 4, Fig 5 and Table 3 show the performance of DTC-HPT classifier model for the chosen metrics. The model accuracy is 99.10%. A performance boost of 50% could be observed after the tuning of hyperparameters. This shows the efficiency of parameter tuning in decision tree. Further, the data imputation using One-hot coding also augmented the attained accuracy.

Table 4: Feature Importance of Clinical Data Columns (DTC)

Clinical Data Columns	Feature Importance
Visit	2.69%
Gender	0.00%
Hand	16.05%
Age	6.53%
SES	6.33%
MMSE	5.92%
eTIV	19.64%
nWBV	15.55%
ASF	0.00%
EDUC	1.04%
MR Delay	5.55%
CDR	20.69%

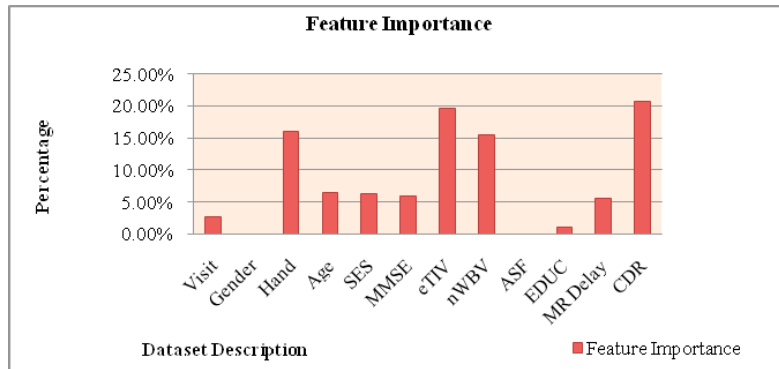
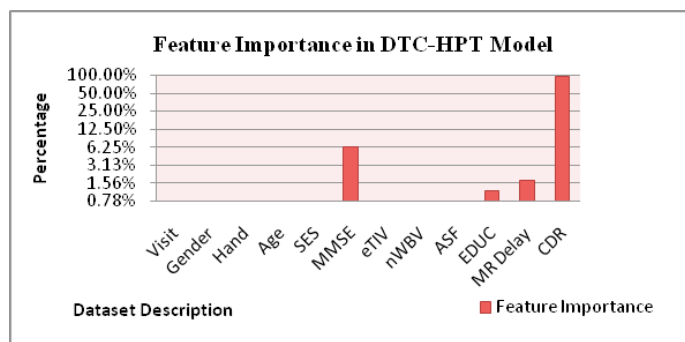
**Fig.6: Feature Importance**

Table 4 and Fig. 6 show the feature importance of Decision Tree Classifier (DTC), prior to the modification prescribed in DTC-HPT. and Table 5 and Fig 7 shows the feature importance of DTC-HPT model.

Table 5: Feature Importance of ClinicalData Columns

Clinical Data Columns	Feature Importance
Visit	0.00%
Gender	0.00%
Hand	0.00%
Age	0.00%
SES	0.00%
MMSE	6.33%
eTIV	0.00%
Nwbv	0.00%
ASF	0.00%
EDUC	1.15%
MR Delay	1.69%
CDR	97.16%

**Fig. 7: Feature Importance in DTC-HPT Model**

Support metric quantifies the number of images considered by DTC during classification against the total number of image count in the experimented data. These results are given in Table 2 and 3 for DTC and DTC_HPT, respectively. The same statistics is visualized in Fig. 3 and 4. The micro average denotes the accuracy of true positive (Precision), and the macro average is a measure of accuracy on false positive (Recall). In the OASIS dataset with 448 data, 112 data are of Mild AD, 112 data are Moderate AD, 112 data are Non-Demented and 112 data are Very Mild AD-affected ones. The DTC-HTP model is accurately supported by all the data in the clinical dataset as depicted by the observed support values

It is apparent from the experimental results that DTC yields accuracy as 41.96%, which has substantially got improved to 99.10% after data imputation and hyper parameter tuning in the proposed DTC-HPT. It is inferred that the modified decision tree exhibits a ~50% increase in performance, in terms of Accuracy. This aspect vouches the efficiency of the newly devised DTC-HPT.

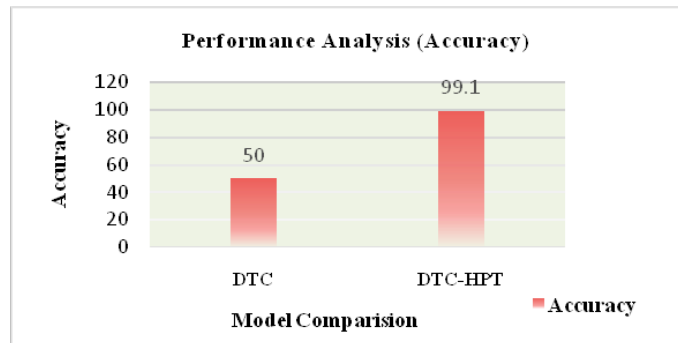


Fig 8: Performance Analysis (Accuracy)

Fig 8 shows the comparison of Accuracy of the two methods: Decision Tree Classifier (DTC) and Decision Tree Classifier after tuning hyper parameter (DTC-HPT) on the imputed data. It could be observed that the modified decision tree exhibits a ~50% increase in performance. This shows that the proposed model has high prediction efficiency.

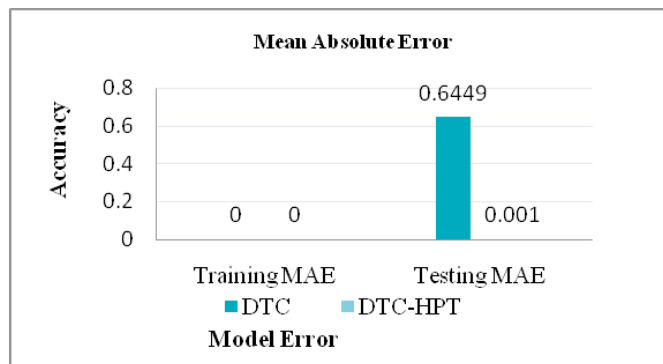


Fig 9: Mean Absolute Error

A comparison analysis for Mean Absolute Error in Fig.9 also confirms the significant improvement in the performance levels between DTC and DTC-HPT. The performance improvement is explicit that endorses the efficiency of the model and its application in the classification and diagnosis of AD.

V.CONCLUSION

Rather than discovering a cure, more attention is paid to the decrease of bottlenecks involved in early intervention and precise diagnosis of dementia. Early detection of AD using different machine learning algorithms and micro simulation methods is elucidated in the literature study. However, the identification of appropriate symptoms which could assist to identify dementia at an early stage remains as a challenge. Decision Tree Classification with hyper parameter tuning (DTC-HPT) technique is a new methodology proposed in this article for AD diagnosis at early and other stages. The decision design is optimized using Entropy and Information Gain. The decision tree is constructed from the root node that descends down to a leaf node depicting the class label for AD detection, based on the ascribe values relating to the new patients, at each splitting node. This proposed work of DTC-HPT technique assures 99.10% of accuracy in AD detection. This research work has ample scope for furtherance to refine the decision tree and thereby increase the precision of AD detection. This proposed machine model increases its clinical utility, as a supplementary validation tool for the human-visual interpretation of MR brain images in the overall management of AD patients. Moreover, this technique can also be applied equally and effectively in other molecular imaging modalities such as PET and Functional MR (fMRI) imaging.

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